MD SAIF ALVI

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Education

Khulna University of Engineering and Technology *Khulna, Bangladesh*
BS, Electronics and Communication Engineering *Jan 2020 – Sep 2025*
GPA: 3.41/4.00 (1 year delay due to Covid)

Research Experience

Undergraduate Thesis 2025
Dept. of Electronics and Communication Engineering, Khulna University of Engineering and Technology
Advisor: Md. Khorshed Alom
External advisor: Dr. Monir Hossen

- Demonstrated that integrating DBSCAN clustering, synchronized dual-side GRU prediction, and sleep-awake scheduling can extend the network lifetime of Wireless Sensor Networks (WSNs) to over 20,000 rounds while achieving an 82.24% transmission avoidance ratio and a 100% Packet Delivery Ratio (PDR).
- Examined the framework using network stability metrics (First Node Death, Half Node Death, Last Node Death), total energy consumption, average latency, and Transmission Avoidance Ratio (TAR) on a simulated 100×100 m homogeneous sensor field against LSTM, LMS-LSTM, and un-scheduled GRU benchmarks.
- Prepared materials for a peer-reviewed conference manuscript published in the 2026 International Conference on Computing, Communication, Security and Intelligent Systems (IC3SIS).

Publications

- **MS Alvi,** RT Joita, MK Alom, "DGS-WSN: A DBSCAN Enabled GRU Based Sleep Scheduling Framework for Energy Efficient Data Transmission in WSNs," IEEE International Conference on Computing, Communication, Security and Intelligent Systems (IC3SIS), Kerala, India, 2026, doi: 10.1109/IC3SIS69949.2026.11608327.
- MAK Kausik, TA Refat, **MS Alvi,** TB Ovi, MA Kabir, "Toward Zero Missed Detections in Low-Data Industrial Fastener Inspection: A Comparative Study of YOLOv11 and YOLOv12," submitted to International Journal of Engineering Science. (*Under review*)

Professional Experience

Software Engineer Jul 2025 - Present
Apploye Inc., Dhaka, Bangladesh
Linux, Shell, Django, Typescript, REST, GraphQL, React, Javascript, PostgreSQL, Payload CMS, Next.js, Shopify, RabbitMQ, BullMQ

- Led full-stack engineering of a lead-collection platform spanning four codebases, modernizing its payment infrastructure, third-party integrations, and UI while the user base increased by 80%.
- Redesigned dashboard APIs, Payment webhooks and system components to improve architecture and efficiency, achieving at least 30% lower API response times.
- Developed a Shopify application, from requirements analysis and system architecture to database design and full-stack implementation, including a fully customizable canvas and 10+ third-party integrations.
- Architected asynchronous worker infrastructure using BullMQ for integrations, scheduling, and background processing, establishing reliable execution of long-running and event-driven tasks.
- Built two production websites from the ground up using Payload CMS, establishing reusable content-management workflows that enabled non-technical teams to manage content independently.

Apprenticeship Dec 2024 - Apr 2025
Geeky Solutions, Remote
Django REST Framework, React, PostgreSQL, GitHub Actions, Pytest

- Under the supervision of a senior engineer, collaborated in a 3-member team to design and implement a full-stack learning management system.
- Led the development of a timed assessment system with dynamic question generation, result analytics, and grading logic.
- Achieved a SonarQube code quality score of 96/100 (3rd highest among 292 teams) and configured the CI/CD pipeline successfully.
- Implemented unit tests, validated APIs and documented the API endpoints to improve reliability.

Selected Projects

- KUETx: Learning Management System. 2025
  Hackathon, Collaborated in a team of 3 members. GitHub
- Wireless Digital Communication System Design under Noise. (MATLAB Simulink) 2024
- Campus-Scale Multi-VLAN Network. (CISCO Packet Tracer) 2024
- Arduino–nRF24L01+PA+LNA based Walkie-Talkie Prototype.
  Advisor: Dr. Monir Hossen 2023
- Inset-Fed Triangular Patch Antenna. (CST Suite) 2023
- FIR-Based Audio Noise Filter. (MATLAB) 2023
- Full-Stack E-commerce Platform Development. (PHP) 2023
- Employee Management System. (Oracle DBMS) 2022
- Regulated DC Power Supply Prototype. 2022

Achievements

- Runner-up, Learnathon 3.0 by Geeky Solutions. 2025
  2nd position among 292 registered teams, 3rd highest code quality score.
- Solved 1000+ algorithmic problems in C++. 2020-2025
  Platforms: Codeforces, AtCoder, and LeetCode.
- Technical scholarship: Merit based award. 2021-2025
  For 4 consecutive academic years.
- 2nd runner-up (Tech), AI hackathon, Apploye Inc. 2025

References

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Assistant Professor
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